

Lab Sheet 11

Title: Working with Indexes.

Objective:

- Adding a Single Field Index
- Understanding Index Restriction
- Creating Compound Indexes
- Using Indexes for Sorting
- Understanding the default index.
- Understanding the text index.

Create an Index

```
db.collection_Name.createIndex ({field name: 1 or -1})
```

Finding the indexes in a collection

```
db.Collection_Name.getIndexes ()
```

Drop indexes in a collection

Dropping a specific index

```
db.Collection_Name.dropIndex ({ index_Name : 1})
```

Dropping all the indexes

```
db.Collection_Name.dropIndex ()
```

Exercise 01 :

Use **persons** database and **personDetails** collection to continue the exercise.

```
// Import JSON File (Run in CMD)
mongoimport --db persons --collection personDetails --file persons.json --jsonArray

// Select Database
use persons

// Display all documents
db.personDetails.find()
```

1. Display the persons whose age is greater than 60.

```
db.personDetails.find({"dob.age": { $gt: 60 }})
```

2. Count the persons whose age is greater than 60.

```
db.personDetails.countDocuments({"dob.age": { $gt: 60 }})
```

3. Analysis the 2nd question.

```
db.personDetails.find({"dob.age": { $gt: 60 }}).explain("executionStats")
```

4. Create an index to age field.

```
db.personDetails.createIndex({"dob.age": })
```

5. Find the details of output.

```
db.personDetails.getIndexes()  
  
// Analyze again after creating index  
db.personDetails.find({"dob.age": { $gt: 60 }}).explain("executionStats")
```

6. Drop an age index

```
db.personDetails.dropIndex({"dob.age": 1})  
  
// Verify indexes  
db.personDetails.getIndexes()  
  
// Create Compound Index  
db.personDetails.createIndex({gender: 1,"dob.age": 1})  
  
// View indexes  
db.personDetails.getIndexes()  
  
// Sort by age ascending  
db.personDetails.find().sort({"dob.age": 1})  
  
// Sort by age descending  
db.personDetails.find().sort({"dob.age": -1})  
  
// Create Text Index  
db.personDetails.createIndex({"name.first": "text","name.last": "text"})  
  
// Search using Text Index  
db.personDetails.find({$text:{$search: "victor"}})
```

```
// View all indexes
db.personDetails.getIndexes()

// Drop all user-created indexes
db.personDetails.dropIndexes()

// Final index check
db.personDetails.getIndexes()
```

Exercise 02:

DB → companyDB

Collection → employeeDetails

1. Import Data using mongoimport

```
// Import JSON File (Run in CMD)
mongoimport --db companyDB --collection employeeDetails --file employeeDetails.json --
jsonArray

// Select Database
use companyDB

// Display all documents
db.employeeDetails.find()
```

2. Display all employees whose age is greater than 60.

```
db.employeeDetails.find({age: { $gt: 60 }})
```

3. Count how many employees have salary greater than 100000.

```
db.employeeDetails.countDocuments({salary: { $gt: 100000 }})
```

4. Display employees from the IT department only.

```
db.employeeDetails.find({department: "IT"})
```

5. Show employees sorted by salary in descending order.

```
db.employeeDetails.find().sort({salary: -1})
```

6. Create a single field index on age.

```
db.employeeDetails.createIndex({age: 1})
```

7. Create a single field index on salary.

```
db.employeeDetails.getIndexes()
```

8. Display all indexes in the collection.

```
db.employeeDetails.find({age: { $gt: 60 }}).explain("executionStats")
```

9. Analyze query performance of:

```
db.employeeDetails.find({salary: { $gt: 100000 }}).explain("executionStats")  
  
// Verify indexes again  
db.employeeDetails.getIndexes()  
  
// Drop age index  
db.employeeDetails.dropIndex({age: 1})  
  
// Drop salary index  
db.employeeDetails.dropIndex({salary: 1})  
  
// Final index check  
db.employeeDetails.getIndexes()
```